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$$\int_0^1 \sqrt{1 - e^{-2x}} dx$$

Neem:

$$u = e^{-x}$$

$$du = -e^{-x} dx$$

$$du = -u dx$$

$$dx = -\frac{du}{u}$$

Grenzen aanpassen:

$$x = 0 \Rightarrow u = 1$$

$$x = 1 \Rightarrow u = e^{-1}$$

Dus:

$$\begin{aligned} & \int_0^1 \sqrt{1 - e^{-2x}} dx \\ &= \int_1^{e^{-1}} \sqrt{1 - u^2} \left(-\frac{du}{u} \right) \\ &= \int_{e^{-1}}^1 \frac{\sqrt{1 - u^2}}{u} du \end{aligned}$$

Neem:

$$u = \cos t$$

$$du = -\sin t dt$$

$$\sqrt{1 - u^2} = \sqrt{1 - \cos^2 t}$$

$$\sqrt{1 - u^2} = \sin t$$

Dan:

$$\begin{aligned} & \int \frac{\sqrt{1 - u^2}}{u} du \\ &= \int \frac{\sin t}{\cos t} (-\sin t) dt \end{aligned}$$

$$\begin{aligned}
&= - \int \frac{\sin^2 t}{\cos t} dt \\
&= - \int \frac{1 - \cos^2 t}{\cos t} dt \\
&= - \int \left(\frac{1}{\cos t} - \cos t \right) dt \\
&= - \int (\sec t - \cos t) dt \\
&= - \ln |\sec t + \tan t| + \sin t
\end{aligned}$$

Terug substitueren:

$$\sin t = \sqrt{1 - u^2}$$

$$\sec t = \frac{1}{u}$$

$$\tan t = \frac{\sqrt{1 - u^2}}{u}$$

$$\sec t + \tan t = \frac{1 + \sqrt{1 - u^2}}{u}$$

Dus:

$$\begin{aligned}
&\int \frac{\sqrt{1 - u^2}}{u} du \\
&= \sqrt{1 - u^2} - \ln \left| \frac{1 + \sqrt{1 - u^2}}{u} \right|
\end{aligned}$$

Nu de grenzen invullen:

$$\begin{aligned}
&\int_{e^{-1}}^1 \frac{\sqrt{1 - u^2}}{u} du \\
&= \left[\sqrt{1 - u^2} - \ln \left| \frac{1 + \sqrt{1 - u^2}}{u} \right| \right]_{e^{-1}}^1 \\
&= 0 - \left(\sqrt{1 - e^{-2}} - \ln \left(\frac{1 + \sqrt{1 - e^{-2}}}{e^{-1}} \right) \right) \\
&= -\sqrt{1 - e^{-2}} + \ln(e(1 + \sqrt{1 - e^{-2}})) \\
&= -\sqrt{1 - e^{-2}} + \ln(e) + \ln(1 + \sqrt{1 - e^{-2}}) \\
&= 1 + \ln(1 + \sqrt{1 - e^{-2}}) - \sqrt{1 - e^{-2}} \\
&= \ln(e + \sqrt{e^2 - 1}) - \frac{\sqrt{e^2 - 1}}{e}
\end{aligned}$$

$$\ln(e + \sqrt{e^2 - 1}) - \frac{\sqrt{e^2 - 1}}{e}$$

References